



microring modulators.



contents

Performance comparison

- Lateral PN junction.
- Vertical PN junction.
- W-Shaped PN junction.

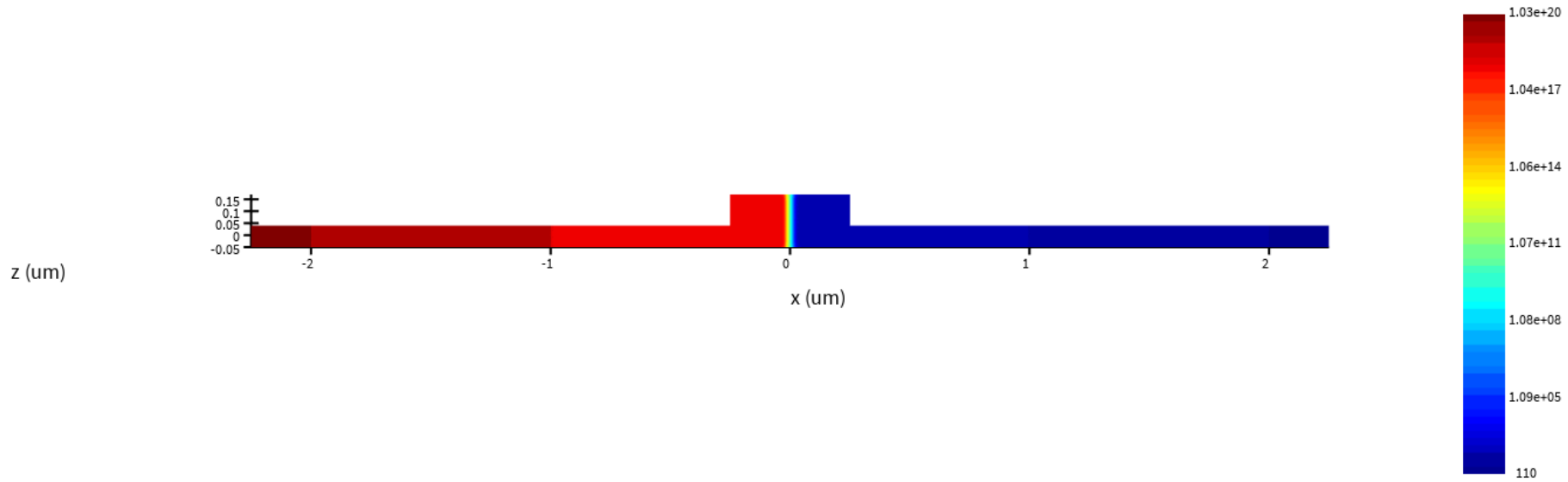
Documentation



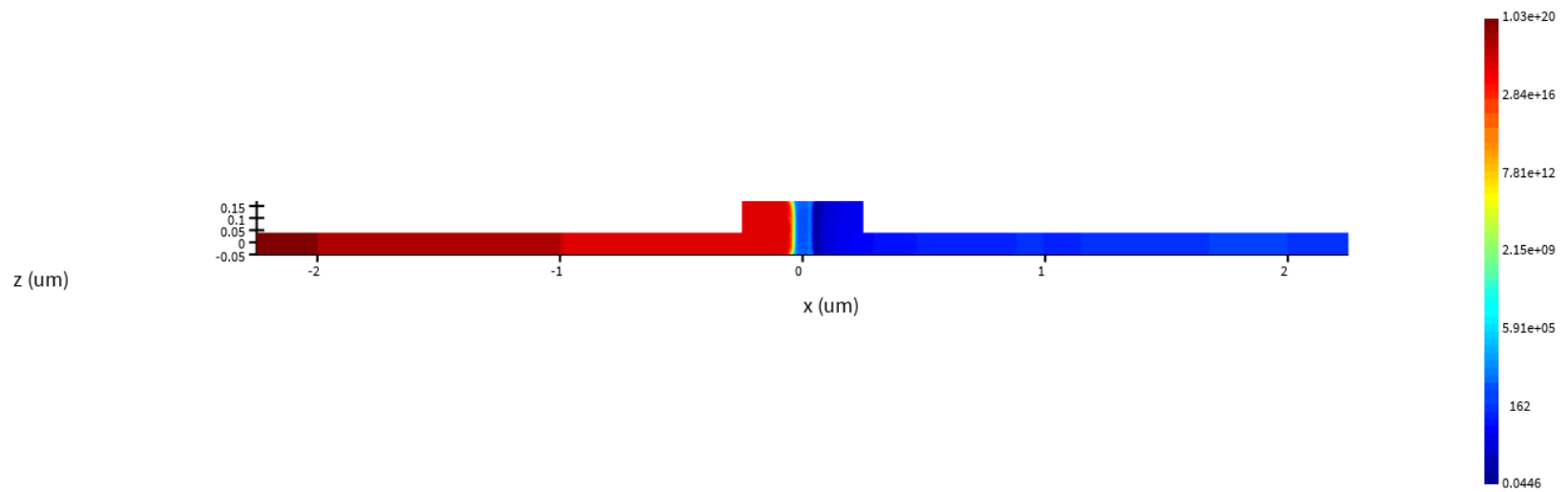
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- ➡ Lateral PN junction.

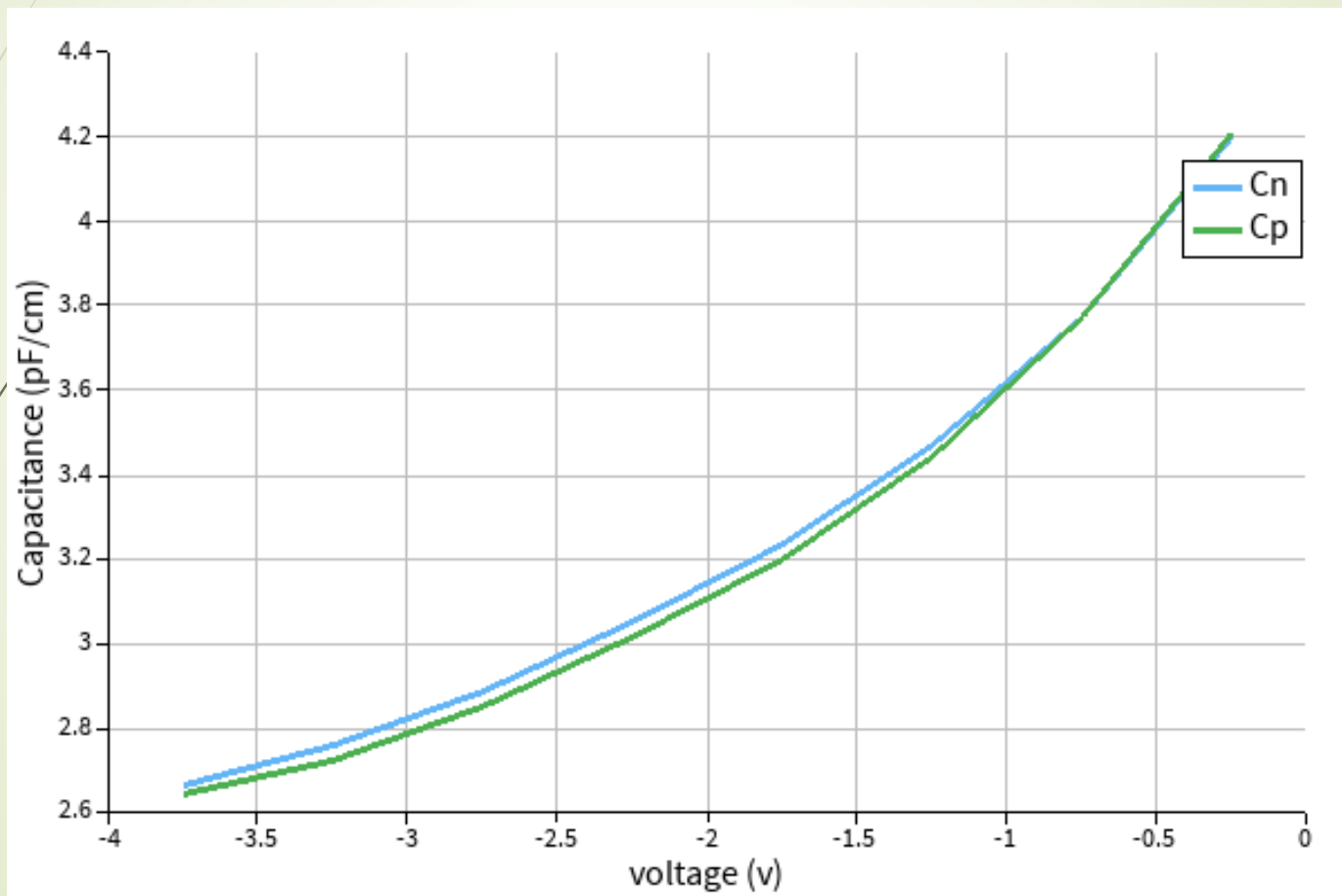
Lateral PN junction



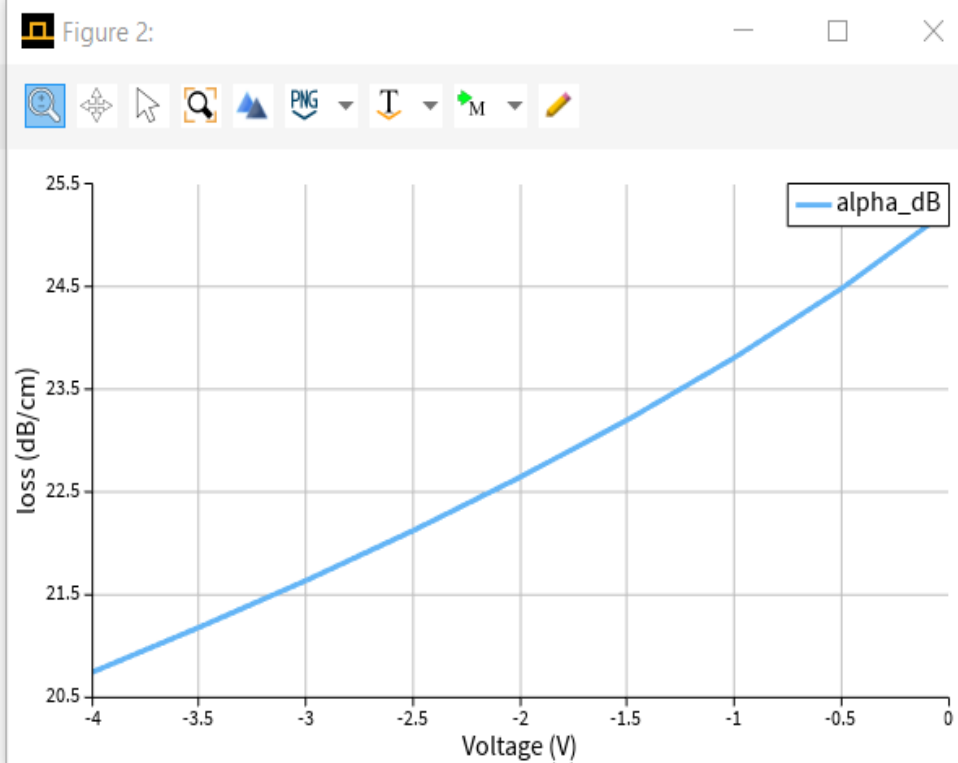
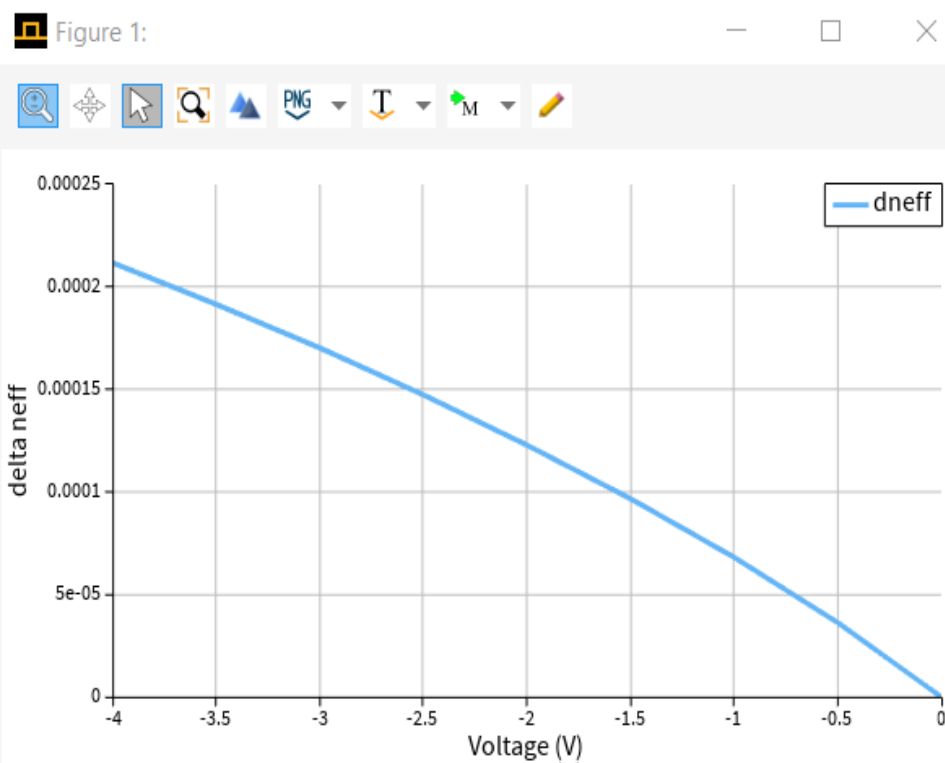
➡ Lateral PN junction



➤ Lateral PN junction



➡ Lateral PN junction





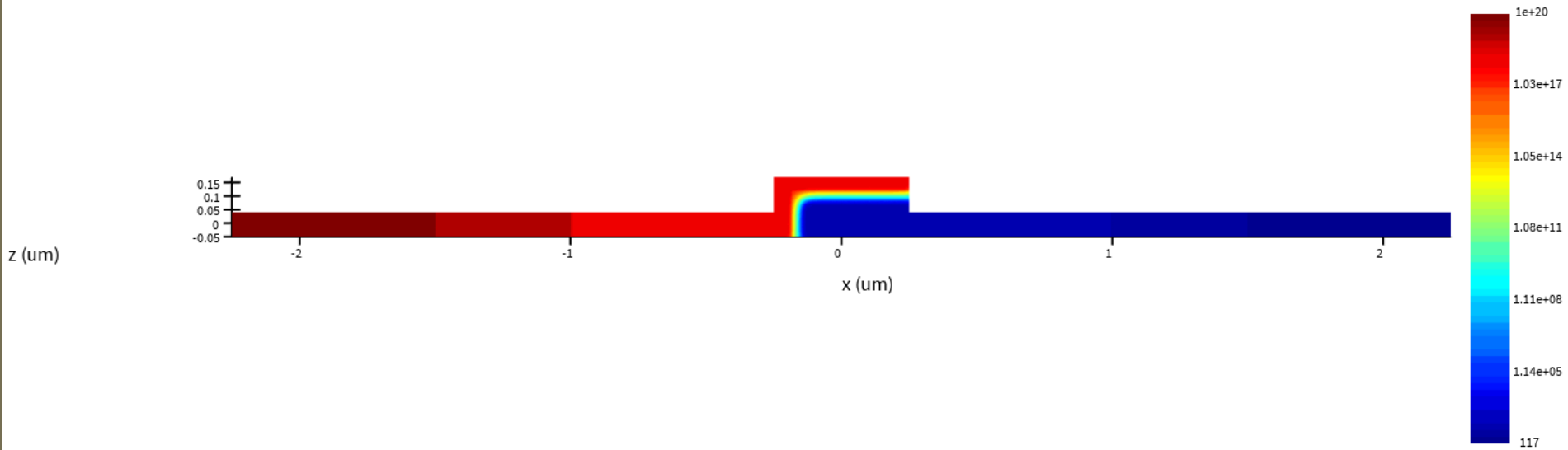
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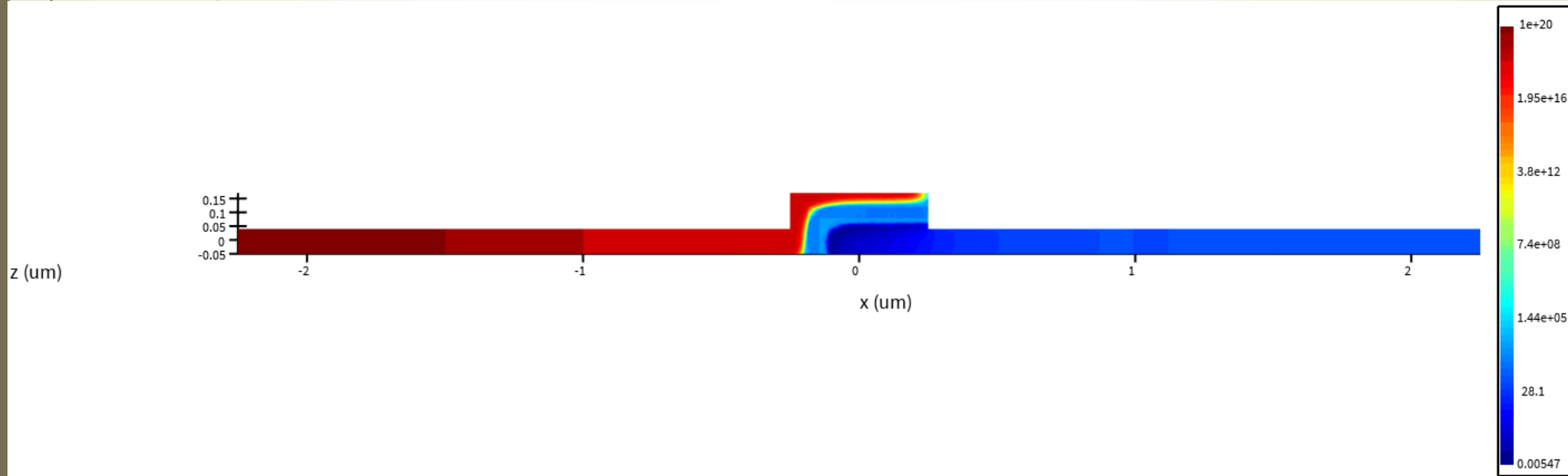
Vertical PN junction.



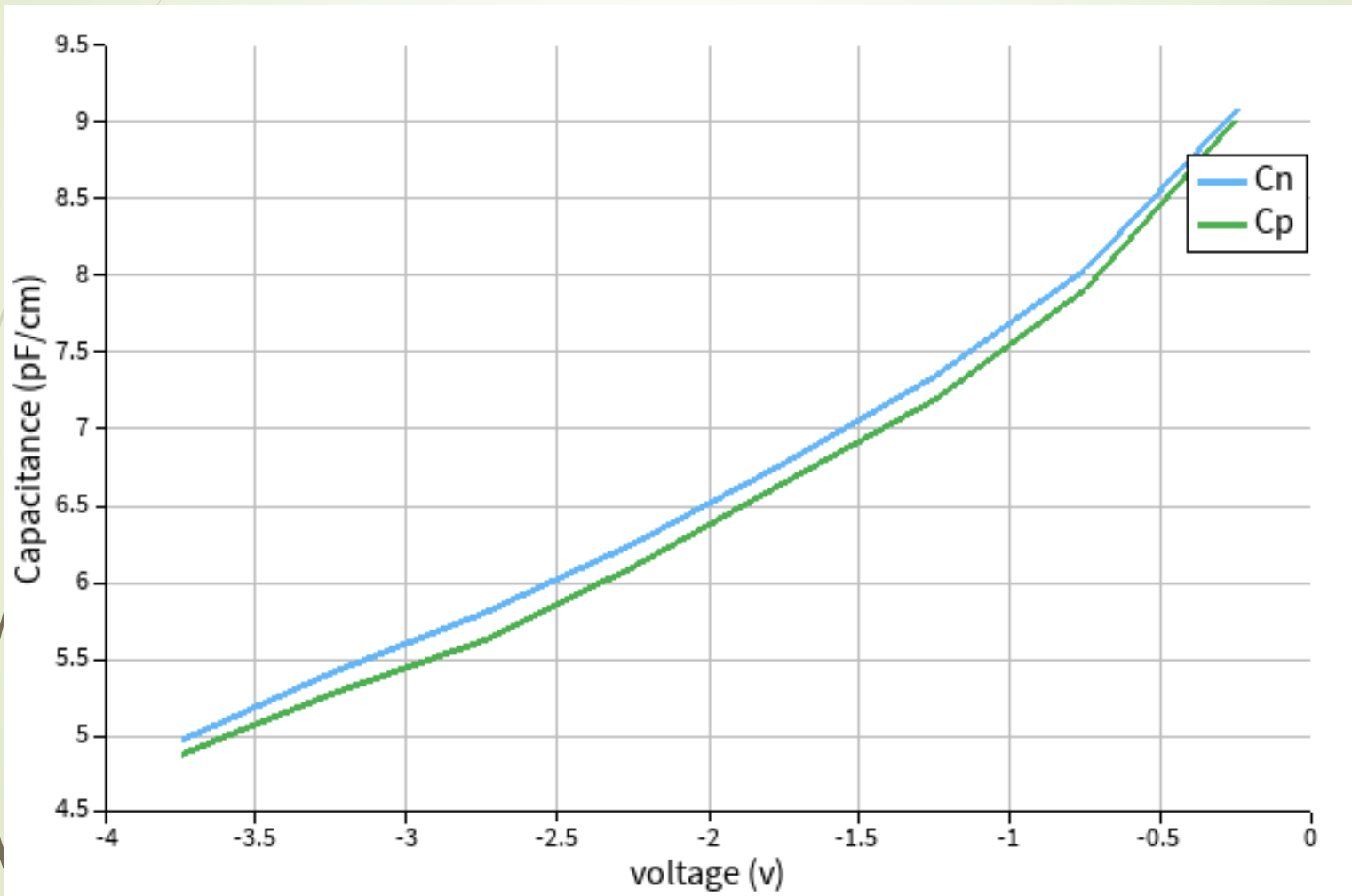
Vertical PN junction simulation



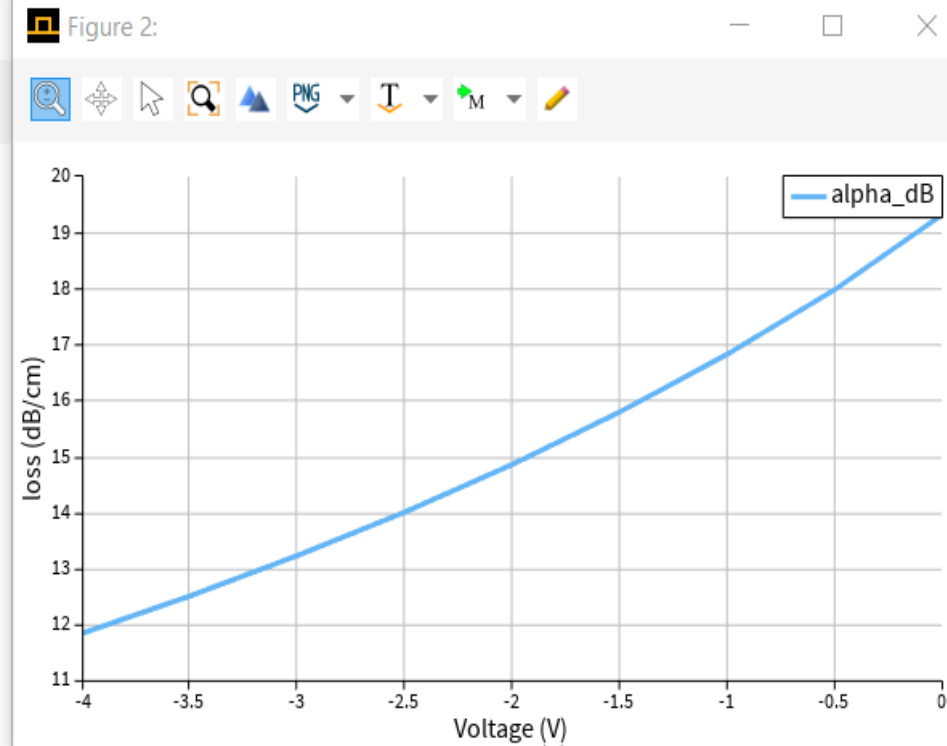
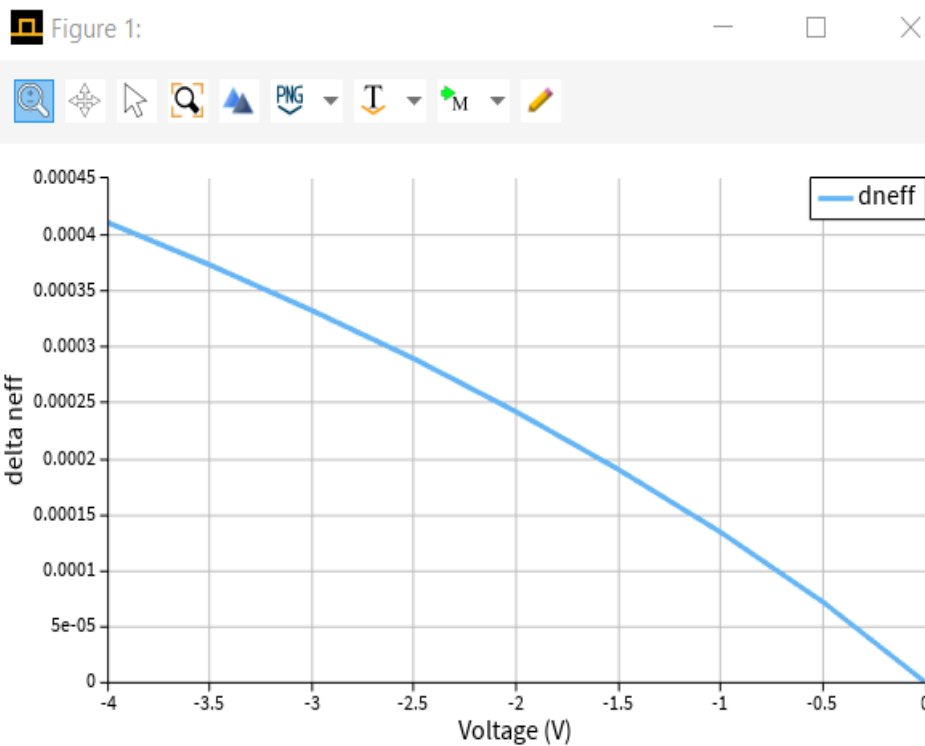
Vertical PN junction simulation



Vertical PN junction simulation



Vertical PN junction simulation





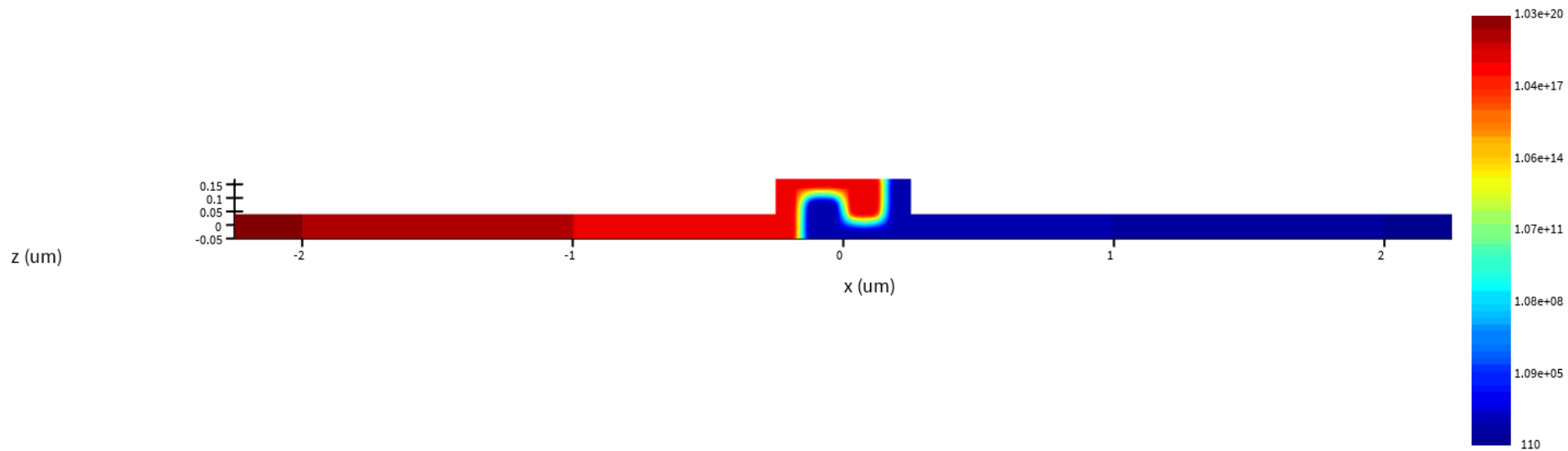
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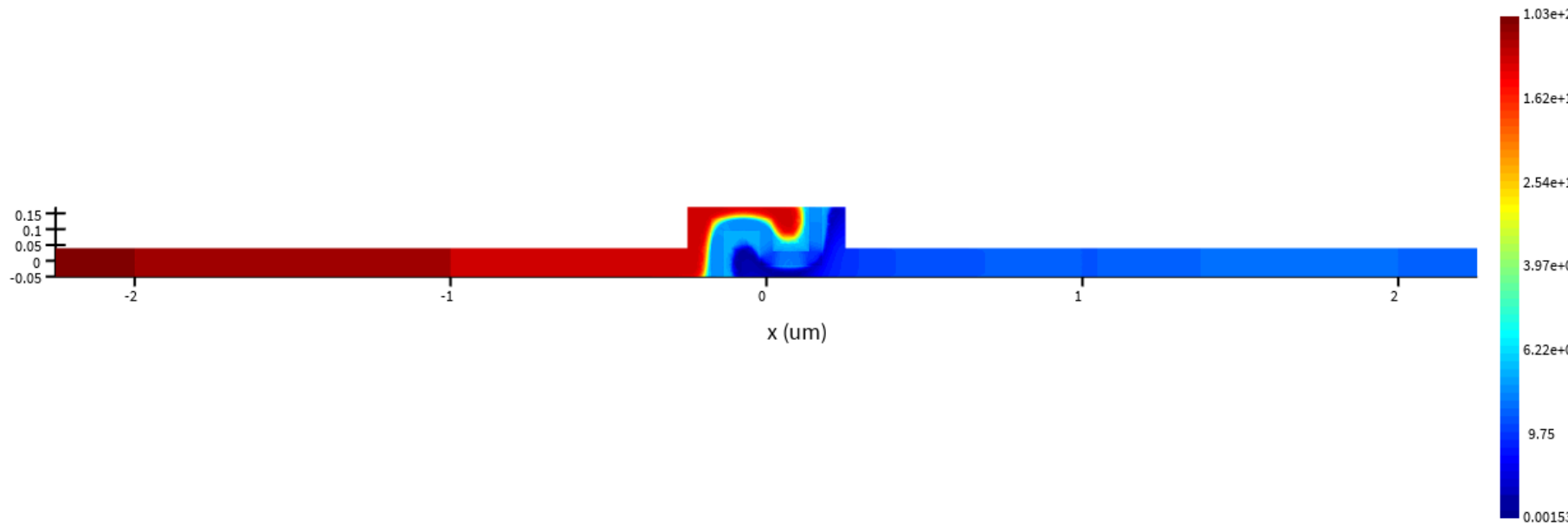
W-Shaped PN junction.



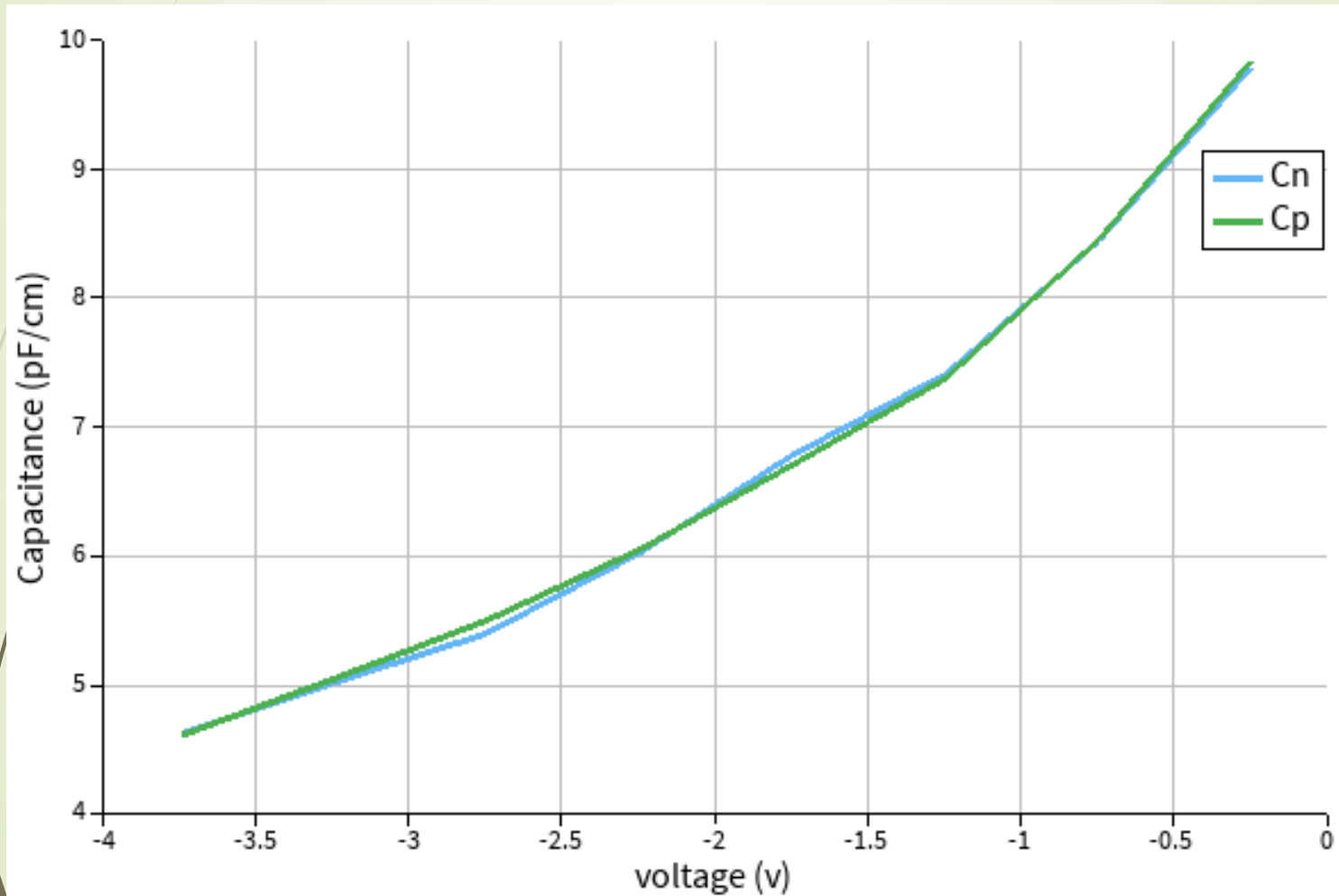
W-Shaped PN junction



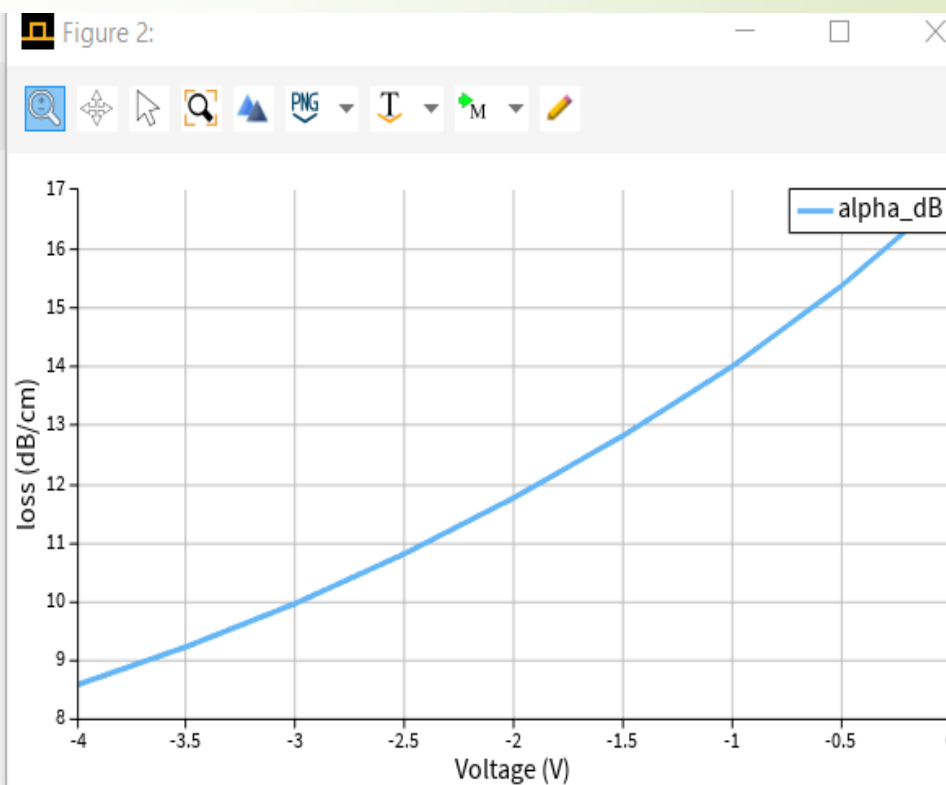
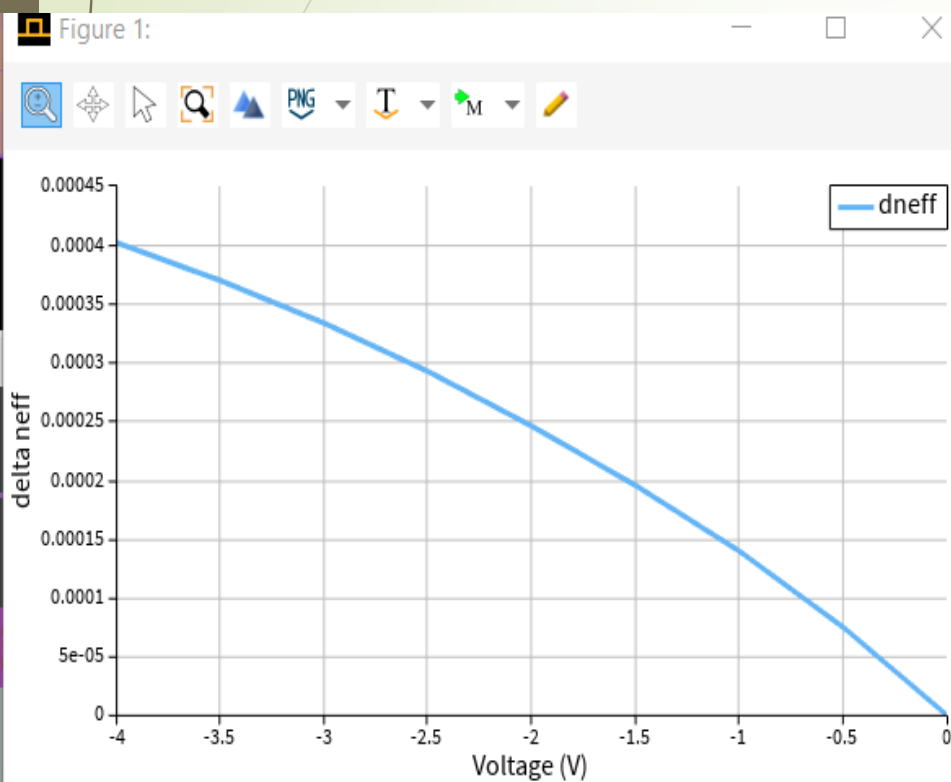
W-Shaped PN junction



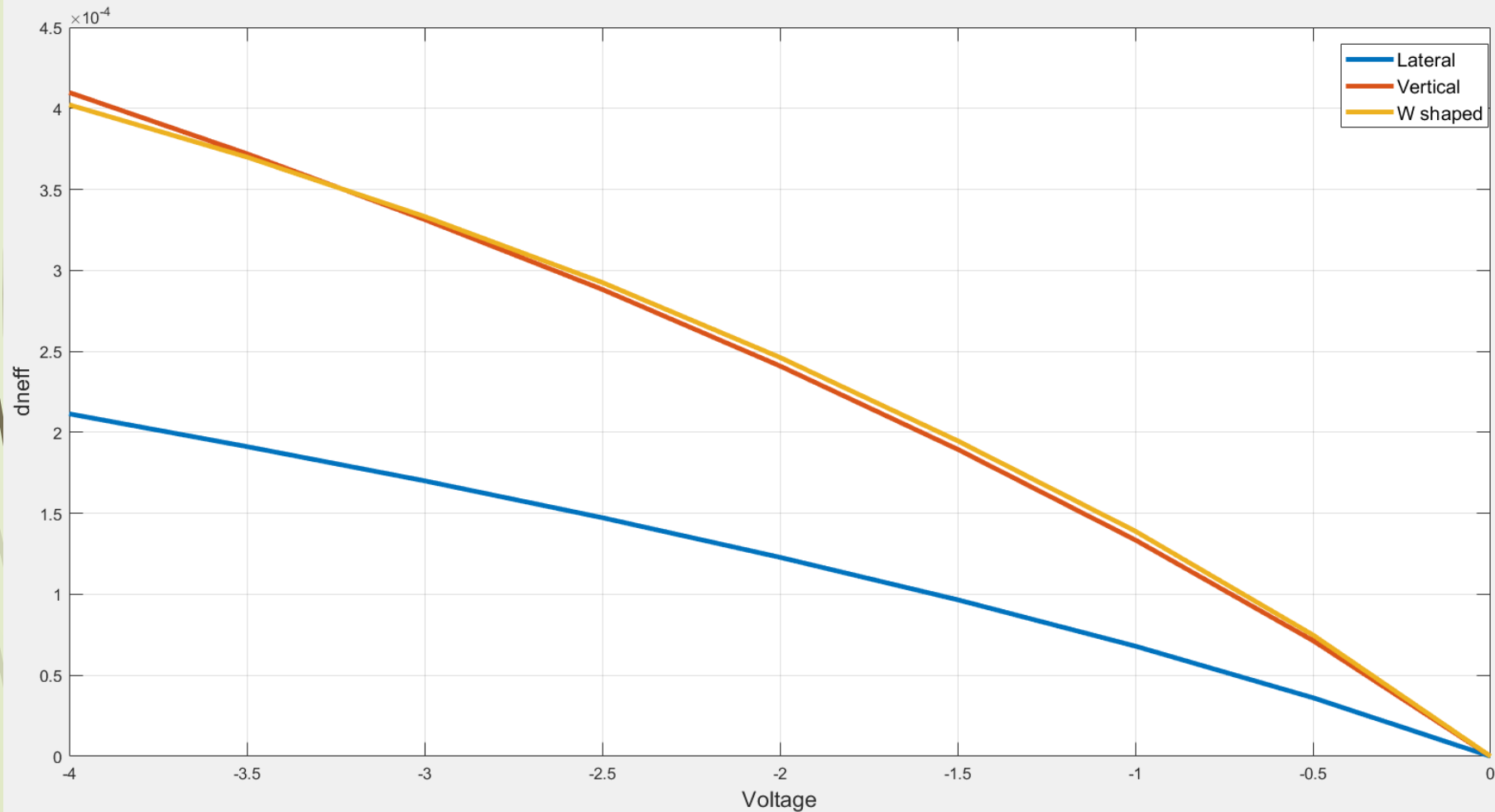
W-Shaped PN junction



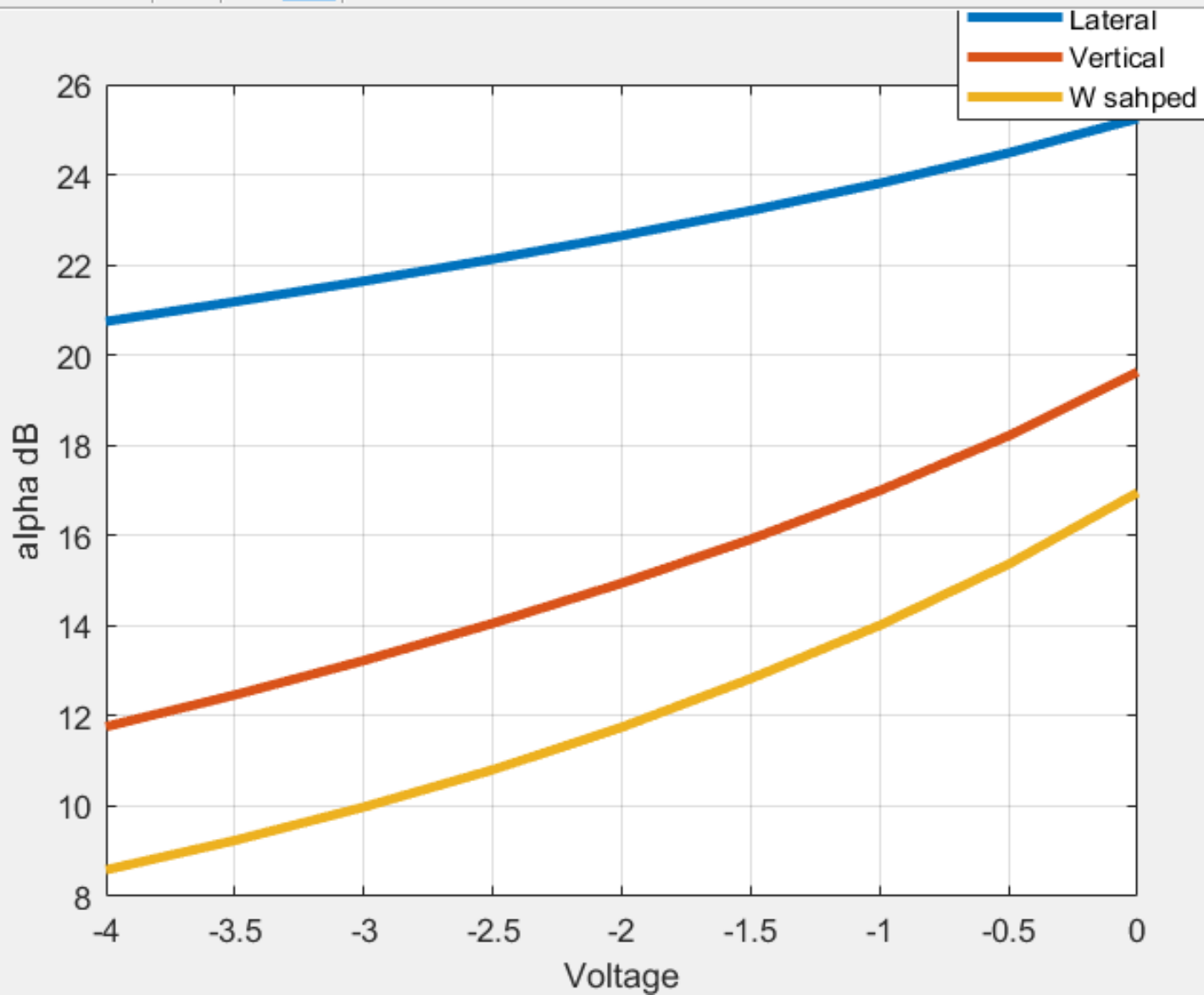
W-Shaped PN junction



junction Comparison

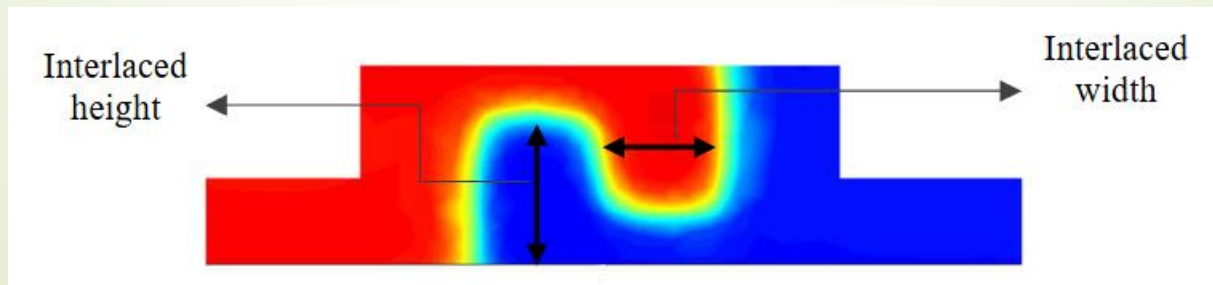
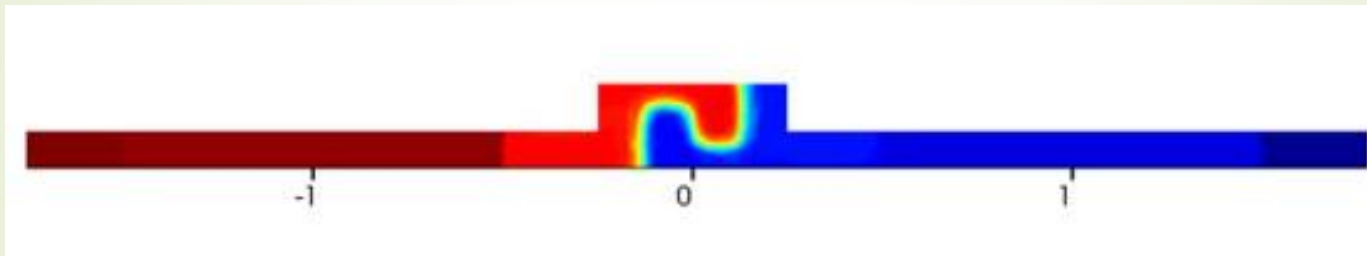


junction Comparison

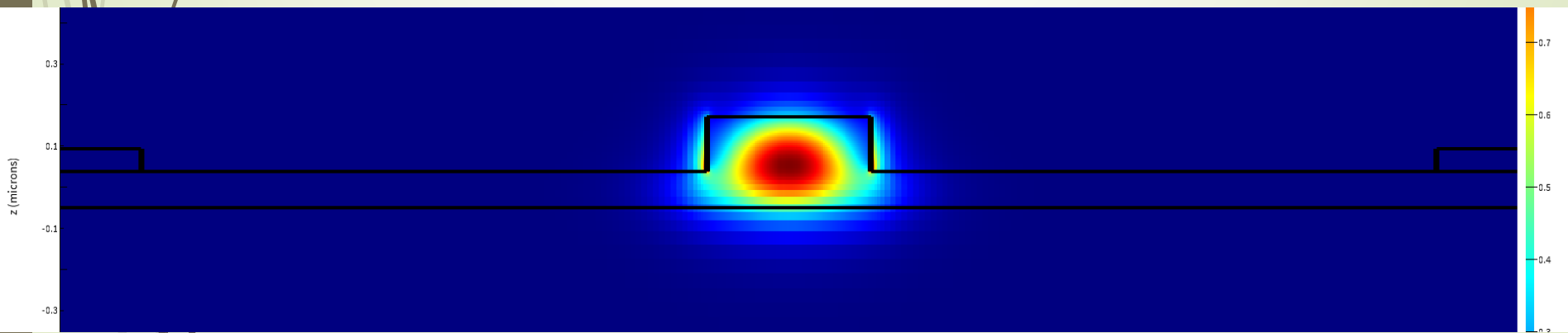
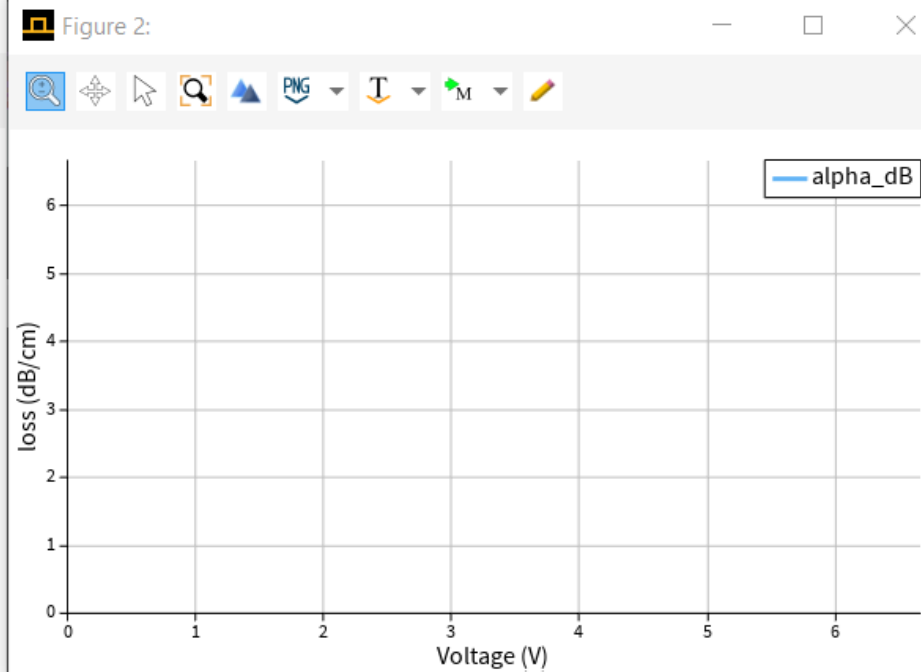
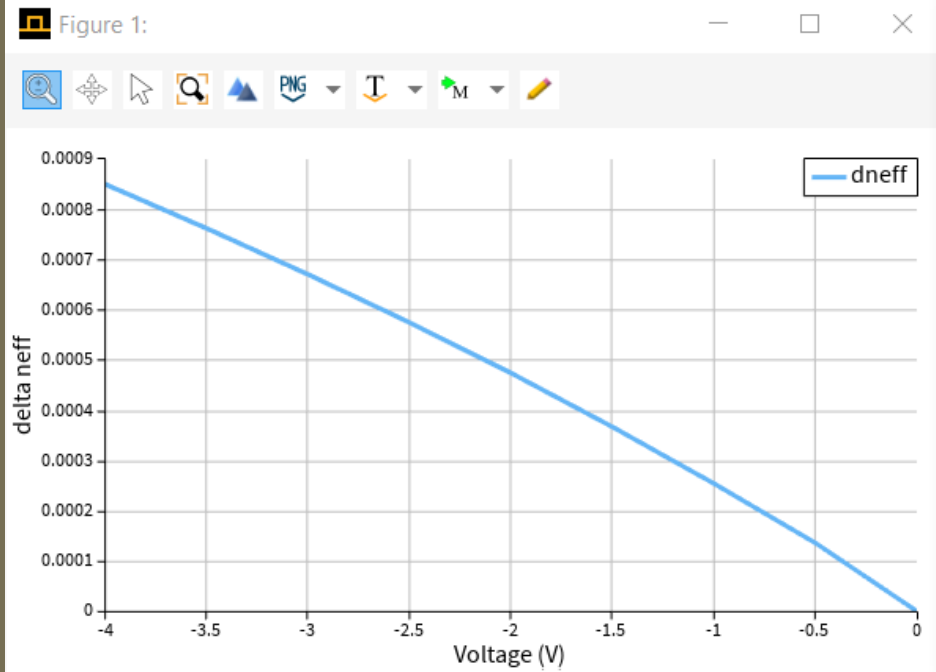


Paper Simulation

- M. Hamouda, C. Mankarious, A. El-Dahshan, A. Fathy, H. Kotb, E. El-Fiky, and D. Khalil, "High efficiency PAM4 ring modulator using W-Shaped PN junction," in *Optical Interconnects and Packaging 2025*, R. Chen and H. Schröder, Eds., Proc. SPIE, vol. 13372, 133720X, SPIE, 2025.



Paper Simulation

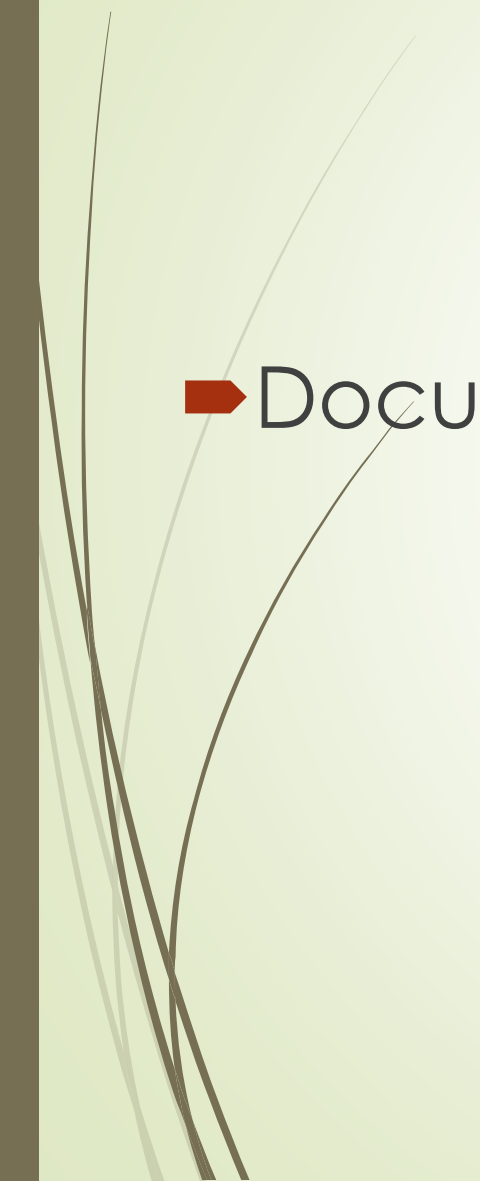


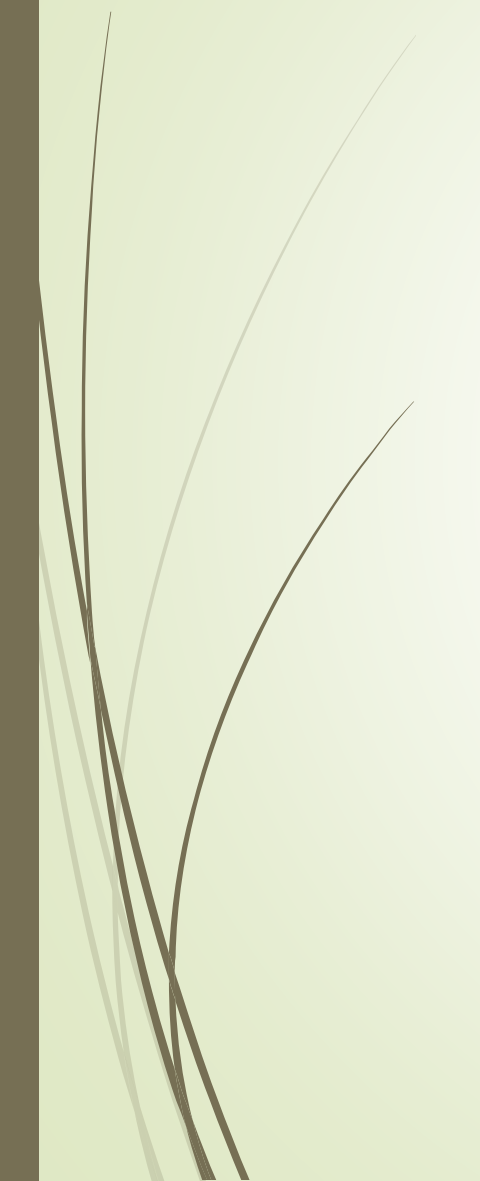


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Documentation.





Thanks 😊